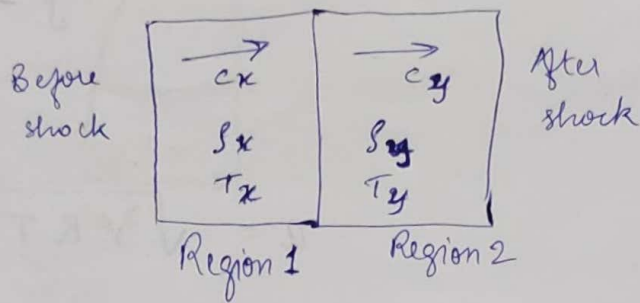


Prandtl-Meyer Relation for a Normal shock



Using energy equation,

$$\underline{h_x} + \frac{1}{2} c_x^2 = h_y + \frac{1}{2} c_y^2 = h + \frac{1}{2} c^{*2}$$

now $h = C_p T$ and $C_p = \frac{\gamma R}{\gamma - 1}$,

$$C_p T_x + \frac{1}{2} c_x^2 = C_p T_y + \frac{1}{2} c_y^2 = C_p T^* + \frac{1}{2} c^{*2}$$

$$\frac{\gamma R T_x}{\gamma - 1} + \frac{1}{2} c_x^2 = \frac{\gamma R T_y}{\gamma - 1} + \frac{1}{2} c_y^2 = \frac{\gamma R T^*}{\gamma - 1} + \frac{1}{2} c^{*2}$$

Now $a = \sqrt{\gamma R T}$

$$\frac{a_x^2}{\gamma - 1} + \frac{1}{2} c_x^2 = \frac{a_y^2}{\gamma - 1} + \frac{1}{2} c_y^2 = \frac{a^{*2}}{\gamma - 1} + \frac{1}{2} a^{*2}$$

$$\frac{a_x^2}{\gamma - 1} + \frac{1}{2} c_x^2 = \frac{a_y^2}{\gamma - 1} + \frac{1}{2} c_y^2 = \frac{a^{*2}}{2} \left(\frac{\gamma + 1}{\gamma - 1} \right)$$

$$\frac{a_x^2}{\gamma-1} + \frac{1}{2} c_x^2 = \frac{a^*^2}{2} \left(\frac{\gamma+1}{\gamma-1} \right)$$

$$a_x^2 = \frac{a^*^2}{2} (\gamma+1) - \frac{(\gamma-1)}{2} c_x^2$$

divide by c_x

$$\frac{a_x^2}{c_x} = \left(\frac{\gamma+1}{2} \right) \left(\frac{a^*^2}{c_x} \right) - \left(\frac{\gamma-1}{2} \right) c_x \longrightarrow (1)$$

$$\frac{a_y^2}{c_y} = \left(\frac{\gamma+1}{2} \right) \left(\frac{a^*^2}{c_y} \right) - \left(\frac{\gamma-1}{2} \right) c_y \longrightarrow (2)$$

Using momentum eqn,

momentum eqn.

$$P A = \dot{m} c$$

$$P_x + \int_x c_x^2 = P_y + \int_y c_y^2 = \text{const}$$

$$P_x - P_y = \int_y c_y^2 - \int_x c_x^2$$

$$P_x - P_y = \int_y c_y c_y - \int_x c_x c_x$$

$$(\because \int_x c_x = \int_y c_y)$$

$$\frac{P_x}{\int_x c_x} - \frac{P_y}{\int_y c_y} = c_y - c_x$$

multiply by γ

$$\frac{\gamma P_x}{\int_x c_x} - \frac{\gamma P_y}{\int_y c_y} = \gamma (c_y - c_x)$$

$$P = \rho R T$$

$$\frac{P}{\rho} = R T$$

$$\frac{\gamma P}{\rho} = \gamma R T = a^2$$

$$a = \sqrt{\gamma R T}$$

$$a^2 = \gamma R T$$

$$\frac{a_x^2}{c_x} - \frac{a_y^2}{c_y} = \gamma (c_y - c_x) \longrightarrow (3)$$

sub eqn ① & ② in ③

$$\left[\left(\frac{\gamma+1}{2} \right) \frac{a^{*2}}{c_x} - \left(\frac{\gamma-1}{2} \right) c_x \right] - \left[\left(\frac{\gamma+1}{2} \right) \frac{a^{*2}}{c_y} - \left(\frac{\gamma-1}{2} \right) c_y \right] = \gamma (c_y - c_x)$$

$$\left(\frac{\gamma+1}{2} \right) a^{*2} \left[\frac{1}{c_x} - \frac{1}{c_y} \right] + \left(\frac{\gamma-1}{2} \right) [c_y - c_x] = \gamma (c_y - c_x)$$

$$\left(\frac{\gamma+1}{2} \right) a^{*2} \left(\frac{c_y - c_x}{c_x \cdot c_y} \right) + \left(\frac{\gamma-1}{2} \right) (c_y - c_x) = \gamma (c_y - c_x)$$

$$\left(\frac{\gamma+1}{2} \right) a^{*2} \cdot \frac{1}{c_x c_y} + \left(\frac{\gamma-1}{2} \right) = \gamma$$

$$\left(\frac{\gamma+1}{2} \right) a^{*2} + \left(\frac{\gamma-1}{2} \right) c_x c_y = \gamma c_x c_y$$

$$\left(\frac{\gamma+1}{2} \right) a^{*2} = \gamma c_x c_y - \left(\frac{\gamma-1}{2} \right) c_x c_y$$

$$\left(\frac{\gamma+1}{2} \right) a^{*2} = c_x c_y \left[\gamma - \frac{(\gamma-1)}{2} \right]$$

$$\left(\frac{\gamma+1}{2} \right) a^{*2} = c_x c_y \left(\frac{\gamma+1}{2} \right)$$

$$\boxed{a^{*2} = c_x \cdot c_y}$$

This is Prandtl-Meyer Relation

Also,

$$\frac{C_x \cdot C_y}{a^{*2}} = 1$$

ie, $\frac{C_x}{a^*} \cdot \frac{C_y}{a^*} = 1$

$\therefore$

$$M_x^* \cdot M_y^* = 1$$

Also, another form of
Prandtl Meyer relation.